PRACTICE WORK 1

Subject Area Analysis (University Database)

Topic: Educational Process at the University

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1. Stored Information in the Future Database

The DB is designed to hold information relating to the educational process at the university.

The database is designed to store information about students, academic groups, teachers, departments, subjects, courses, classrooms, class schedules, and students' grades. The database will allow the university to store, organize, update, and retrieve information related to the educational process.

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2. Entities and Attributes

The future database will contain **9 main entities (tables)**.

No.	Entity (Table)	Attributes	Primary Key
1	**Students**	student_id, last_name, first_name, birth_date, gender, phone, email, group_id	**student_id**
2	**Groups**	group_id, group_name, enrollment_year, specialty, department_id	**group_id**
3	**Teachers**	teacher_id, last_name, first_name, academic_degree, phone, email, department_id	**teacher_id**
4	**Departments**	department_id, department_name, faculty_name	**department_id**
5	**Subjects**	subject_id, subject_name, credits, semester, department_id	**subject_id**
6	**Courses**	course_id, subject_id, teacher_id, group_id, academic_year, semester	**course_id**

7	**Classrooms**	classroom_id, building, room_number, capacity, classroom_type	**classroom_id**
8	**Schedule**	schedule_id, course_id, classroom_id, day_of_week, start_time, end_time, lesson_type	**schedule_id**
9	**Grades**	grade_id, student_id, course_id, grade_value, assessment_date, assessment_type	**grade_id**

.For example:

Students (student_id, last_name, first_name, birth_date, gender, phone, email, group_id)

Here, **student_id** is the Primary Key.

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3. Relationships Between Entities

No.	First Entity	Second Entity	Relationship	Description
1	Students	Groups	N:1	One student can be enrolled in only one group, while one group can contain many students.
2	Groups	Departments	N:1	One group belongs to one department, while one department can have several groups.
3	Teachers	Departments	N:1	One teacher works in one department, while one department can have several teachers.
4	Subjects	Departments	N:1	One subject belongs to one department, while one department can have several subjects.
5	Subjects	Courses	1:N	One subject can be offered as several courses, while one course belongs to one subject.
6	Teachers	Courses	1:N	One teacher can teach several courses, while one course is taught by one teacher.
7	Groups	Courses	1:N	One group can attend several courses, while one course is assigned to one group.

8	Courses	Schedule	1:N	One course can have several scheduled lessons, while one scheduled lesson belongs to one course.
9	Classrooms	Schedule	1:N	One classroom can be used for several lessons at different times, while one lesson takes place in one classroom.
10	Students	Grades	1:N	One student can receive several grades, while one grade belongs to one student.
11	Courses	Grades	1:N	One course can have grades for many students, while one grade refers to one course.

Detailed Description of Relationships

Students — Groups:
One student can be enrolled in only one group, while one group can contain many students.

Groups — Departments:
One group belongs to one department, while one department can contain several groups.

Teachers — Departments:
One teacher works in one department, while one department can have many teachers.

Subjects — Departments:
One subject belongs to one department, while one department can be responsible for several subjects.

Subjects — Courses:
One subject can be taught as several courses, while one course refers to only one subject.

Teachers — Courses:
One teacher can teach several courses, while one course is taught by one teacher.

Groups — Courses:
One group can study several courses, while one course is assigned to one group.

Courses — Schedule:
One course can have several scheduled lessons, while each scheduled lesson belongs to one course.

Classrooms — Schedule:
One classroom can be used for several lessons at different times, while one lesson takes place in one classroom.

Students — Grades:
One student can receive several grades, while each grade belongs to one student.

Courses — Grades:
One course can have grades for many students, while each grade refers to one course.

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4. Constraints

No.	Constraint
1	Each student must have a unique `student_id`.
2	Each group must have a unique `group_id`.
3	Each teacher must have a unique `teacher_id`.
4	Each department must have a unique `department_id`.
5	Each subject must have a unique `subject_id`.
6	Each course must have a unique `course_id`.
7	Each classroom must have a unique `classroom_id`.
8	Each schedule record must have a unique `schedule_id`.
9	Each grade must have a unique `grade_id`.
10	A student's date of birth cannot be in the future.
11	One student can belong to only one group at a particular time.
12	The number of students in a group cannot exceed the established group capacity.
13	Classroom capacity must be greater than 0.
14	The number of credits for a subject must be greater than 0.
15	A student's grade must be between 0 and 100.
16	The end time of a lesson must be later than its start time.
17	Each course must refer to an existing subject, teacher, and group.
18	Each schedule record must refer to an existing course and classroom.
19	Each grade must refer to an existing student and course.
20	A student cannot have duplicate grade records for the same course and assessment type.

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5. User Groups and Their Access Rights

The database is designed for university administrators, teachers, students, department staff, and university management.

No.	User Group	Access Rights
1	**Administrator**	Can view, add, modify, and delete records in all tables and manage user access rights.
2	**Teacher**	Can view students enrolled in their courses and enter or modify grades.
3	**Student**	Can view their own schedule, subjects, and grades.
4	**Department Staff**	Can manage information about groups, subjects, and teachers belonging to their department.
5	**University Management**	Can view reports and statistics about students, academic performance, teachers, and subjects.

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6. Potential Questions from Users to the Database (Queries)

1. Which students are enrolled in a particular group?
2. Which group does a particular student belong to?
3. What subjects does a particular group study?
4. Which teacher teaches a particular subject?
5. What courses does a particular teacher teach?
6. What is the schedule of a particular group?
7. In which classroom does a particular lesson take place?
8. What grades has a particular student received?
9. What is the average grade of a particular student?
10. Which students have the highest academic performance?
11. How many students are there in each group?
12. How many teachers work in each department?
13. What subjects are offered by a particular department?
14. Which classrooms are available at a particular time?
15. Which students received low grades in a particular subject?
16. How many courses does each teacher teach?
17. What is the average grade of a particular group?
18. Which group is enrolled in a particular course?
19. Which classroom has the highest capacity?
20. What are the grades of a particular student in all subjects?

Conclusion

This work presents a subject area analysis for a database designed to support the educational process at a university. The database contains **9 main entities (tables): Students, Groups, Teachers, Departments, Subjects, Courses, Classrooms, Schedule, and Grades**.

The attributes and Primary Keys of each entity have been identified. The relationships between the entities, database constraints, user groups and their access rights, and potential database queries have also been defined. This analysis provides a foundation for the further design and implementation of the university educational process database.